ModelIt!

Teacher's Guide

The Plastic Paradox

Modeling Microplastic Bioaccumulation, Ocean Gyres, and Decomposition Timescales

Quick Reference

Grade Level	Grade 11L2
NGSS Standard	HS-ESS3-4, HS-LS2-6
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS3-4, HS-LS2-6

Evaluate or refine a technological solution that reduces impacts of human activities on natural systems. Evaluate claims, evidence, and reasoning that the complex interactions in ecosystems maintain relatively consistent numbers and types of organisms in stable conditions, but changing conditions may result in a new ecosystem.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model of microplastic transport, fragmentation, and bioaccumulation to predict how plastic pollution propagates through ocean circulation and marine food webs over decadal timescales.
Disciplinary Core Idea	**ESS3.C Human Impacts on Earth Systems / LS2.C Ecosystem Dynamics** — Students model how human-generated plastic pollution alters ocean chemistry and disrupts ecosystem dynamics through bioaccumulation and biomagnification across trophic levels.
Crosscutting Concept	**Cause and Effect / Scale, Proportion, and Quantity** — Students analyze how cause-and-effect relationships change across spatial and temporal scales — from molecular fragmentation to ocean-basin circulation to generational ecosystem contamination.

CAST Testing Alignment

MULTIPLE CHOICE:

The Great Pacific Garbage Patch contains an estimated 1.8 trillion plastic pieces weighing 80,000 tons. Current cleanup technology can remove approximately 10,000 tons per year, but 11 million tons of new plastic enter the ocean annually. Based on your model, which statement BEST describes the effectiveness of ocean cleanup as a standalone solution?

CONSTRUCTED RESPONSE:

Using your model, explain why a microplastic particle ingested by a copepod in the open ocean can result in dangerous toxin concentrations in a tuna consumed by a human on a different continent. Reference bioaccumulation, biomagnification, gyre circulation, and at least three trophic levels in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Plastic is the defining material of the modern era — lightweight, durable, cheap, and endlessly versatile. But the same molecular stability that makes plastic so useful means it persists in the environment for centuries to millennia. Understanding the microplastic crisis requires modeling the intersection of materials science, ocean physics, toxicology, and food web ecology to reveal why a convenience material has become a global contamination emergency.

From Convenience to Crisis

Global plastic production has grown from 2 million tons in 1950 to over 400 million tons today — a 200-fold increase in seven decades. Approximately 40% of plastic produced is single-use packaging discarded within minutes of purchase. Of all plastic ever produced, only 9% has been recycled, 12% incinerated, and 79% has accumulated in landfills or the natural environment. An estimated 11 million metric tons of plastic enter the ocean each year through rivers, coastal runoff, and direct dumping. Without systemic change, ocean plastic is projected to triple by 2040.

The Physics of Fragmentation and Gyre Trapping

Large plastic items in the ocean are attacked by UV radiation, which breaks polymer chains through photodegradation, and by wave action, which mechanically fragments weakened material. This process creates an exponentially growing number of progressively smaller particles — one plastic bottle can fragment into millions of microplastic pieces. Ocean gyres — massive circular current systems driven by wind and the Coriolis effect — sweep floating debris toward their calm centers. The five major ocean gyres have created permanent accumulation zones; the Great Pacific Garbage Patch alone covers 1.6 million square kilometers. Critically, microplastics smaller than 1mm are nearly impossible to remove from open water without capturing plankton and other organisms.

Bioaccumulation and Biomagnification

Microplastics enter marine food webs primarily through ingestion by zooplankton, larval fish, and filter feeders that mistake plastic particles for food. Once ingested, microplastics and the persistent organic pollutants (POPs) adsorbed to their surfaces accumulate in fatty tissues. As predators consume many contaminated prey items, toxin concentrations magnify at each trophic level — a process called biomagnification. Studies show that PCB concentrations on microplastic surfaces can be 100,000-1,000,000 times higher than in surrounding seawater because hydrophobic toxins preferentially bind to plastic. A top predator consuming thousands of contaminated fish over its lifetime can accumulate toxin levels millions of times higher than ambient water concentrations.

Human Health and Global Consequences

Microplastics have been detected in human blood, lung tissue, placental tissue, and breast milk. The average person ingests an estimated 5 grams of microplastic per week — roughly the weight of a credit card. Health effects are still being studied, but laboratory research links microplastic exposure to inflammation, endocrine disruption, oxidative stress, and cellular damage. The economic costs are also staggering: marine plastic pollution causes an estimated $13 billion in annual damage to marine ecosystems, fisheries, and tourism. Environmental justice concerns are acute — plastic production facilities and waste disposal sites are disproportionately located near low-income communities and communities of color, while ocean plastic disproportionately affects developing nations dependent on marine resources.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify six components of the microplastic system: Plastic Production and Disposal Rate, UV and Mechanical Fragmentation, Ocean Gyre Concentration, Microplastic Ingestion Rate, Trophic-Level Toxin Concentration, and Marine Ecosystem Health — distinguishing the human-controlled input (production/disposal) from the natural processes that disperse, concentrate, and magnify contamination.

E **Establish Relationships**

Students determine that Plastic Production drives the input of debris, which UV and Mechanical Fragmentation breaks into microplastics. Ocean Gyre Concentration traps and accumulates these particles. Microplastic Ingestion Rate introduces contamination to the food web. Trophic-Level Toxin Concentration magnifies harm at each step. Marine Ecosystem Health degrades as cumulative contamination exceeds organism tolerance.

V **Visualize & Model**

Students build a computational model showing the cascade from production through fragmentation, gyre trapping, food web entry, and biomagnification — visualizing why contamination grows nonlinearly even with constant input.

E **Evaluate Outcomes**

Students run current trajectory, source reduction, and cleanup + reduction scenarios to quantify the lag between intervention and ecosystem recovery and identify which strategies meaningfully reduce contamination within human-relevant timescales.

R **Revise & Extend**

Students add Nanoplastic Formation, Chemical Leaching Rate, or Sediment Burial Rate to explore how sub-visible contamination pathways, chemical exposure beyond ingestion, and deep-ocean sinks create additional dimensions of the crisis.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Every piece of plastic ever made still exists somewhere on this planet. And increasingly, that somewhere is inside living things — including us."*

Watch For

Show a 60-second video of microplastics being filtered from ocean water samples. Display the statistic: you eat about a credit card's worth of plastic every week. Let students react. Ask: Where does it come from?

Slide 2: Learning Objectives

3 min

Say: *"Today you're modeling how a material designed to last forever becomes the most pervasive contaminant on Earth — and why it's almost impossible to clean up."*

Watch For

Have students read objectives. Pre-teach bioaccumulation vs. biomagnification using a stacking demonstration — one block per organism at the base, stack blocks as you move up trophic levels to show concentration multiplying. Quick-write: What's the difference between something that breaks down and something that breaks apart?

Slide 3: Big Question

3 min

Say: *"We invented a miracle material — lightweight, cheap, moldable into anything. The problem? We designed it to last 500 years and we use it for 5 minutes."*

Watch For

Display a timeline: plastic bag used for 12 minutes, persists for 500 years. Quick calculation: what's the ratio of use time to persistence time? (1:22 million). Discuss: what other product has that ratio?

Slide 4: LEVER Framework

2 min

Say: *"We're modeling a one-way pipeline: plastic goes in, nothing comes out. The ocean is the end of the line — and every trophic level is a multiplier."*

Watch For

Preview each LEVER step. Emphasize the concept of timescale mismatch: human decision-making happens in years, plastic persistence spans centuries. The model will reveal why this mismatch makes cleanup nearly impossible.

Slide 5: Activity 1: Components

8 min

Say: *"One human decision. Five natural processes. But here's the paradox — every natural process in this system makes the problem worse, not better."*

Watch For

Guide sorting of external vs. internal components. Only Plastic Production and Disposal Rate is external. Discuss why UV Fragmentation is internal — nature is breaking plastic apart, but that creates more particles, not fewer. Fragmentation is degradation without decomposition.

Slide 6: Activity 2: Connections

8 min

Say: *"Follow the plastic: factory to ocean, sunlight shatters it, gyres trap it, plankton eat it, fish eat the plankton, we eat the fish. At every step, the concentration goes up."*

Watch For

Help students map the cascade. Identify the biomagnification multiplier: if toxin concentration increases 10x at each trophic level and there are 4-5 levels, what's the total magnification? (10,000-100,000x). Find why this system has no natural negative feedback — nothing in the ocean breaks plastic down.

Slide 7: Activity 3: Simulation

10 min

Say: *"Three futures: keep dumping, cut the source in half, or cut the source AND try to clean the ocean. Which one actually saves the food web?"*

Watch For

Students predict before running simulations. Key focus: the lag time between reducing input and seeing ecosystem improvement. Even with 50% reduction, existing microplastics persist for centuries. Compare Trophic-Level Toxin Concentration across scenarios at 2040, 2060, and 2080.

Slide 8: Discoveries

5 min

Say: *"Your model just revealed the plastic paradox: the problem is already so large that cleanup is like emptying the ocean with a teaspoon. The only real solution is turning off the faucet."*

Watch For

Lead discussion on why prevention beats remediation by orders of magnitude. Connect to real-world examples: The Ocean Cleanup project, plastic bag bans, extended producer responsibility laws. Discuss: if cleanup can't solve it, what role does it still play?

Slide 9: STEM Challenge

12 min

Say: *"A fishing industry worth half a billion dollars is being contaminated by plastic. Design the system that stops the contamination — not in a century, but in a generation."*

Watch For

Groups design integrated reduction strategies addressing source, transport, accumulation, and monitoring. Must include material alternatives, waste management infrastructure, cleanup technology, and a verification plan. Present as environmental engineering proposals to a coastal resource board.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Plastic Production and Disposal Rate
- **INTERNAL:** UV and Mechanical Fragmentation, Ocean Gyre Concentration, Microplastic Ingestion Rate, Trophic-Level Toxin Concentration, Marine Ecosystem Health

Relationships

- **Plastic Production to UV and Mechanical Fragmentation:** POSITIVE (+) — More plastic entering the ocean increases the total mass available for photodegradation and mechanical breakdown. Each large item fragments into thousands of microplastic particles, so the relationship amplifies the particle count exponentially relative to mass input.
- **UV and Mechanical Fragmentation to Ocean Gyre Concentration:** POSITIVE (+) — More microplastic particles created by fragmentation increases the material available to be swept into gyre accumulation zones by ocean currents. Smaller particles are more easily transported by currents, so fragmentation actually increases gyre concentration efficiency.
- **Ocean Gyre Concentration to Microplastic Ingestion Rate:** POSITIVE (+) — Higher concentrations of microplastics in the water column increase the probability of ingestion by marine organisms. In gyre centers, plankton-sized plastic particles can outnumber actual plankton by 6:1, making ingestion virtually inevitable for filter feeders.
- **Microplastic Ingestion Rate to Trophic-Level Toxin Concentration:** POSITIVE (+) — Higher ingestion rates at the base of the food web introduce more contaminated organisms into the food chain. Each predator consumes many prey items, concentrating toxins at each trophic level through biomagnification — a 10x multiplier per level.
- **Trophic-Level Toxin Concentration to Marine Ecosystem Health:** NEGATIVE (-) — Higher toxin concentrations in organisms reduce reproductive success, impair immune function, disrupt endocrine systems, and increase mortality — particularly in top predators where concentrations are highest. Population declines cascade through the food web.

Key Discoveries

- Microplastics have been detected in 90% of commercial fish species, in Arctic sea ice, in deep-sea sediments 11,000 meters below the surface, and in human blood and placental tissue — the contamination is already global and pervasive
- Plastic does not biodegrade on human timescales — it photodegrades into ever-smaller fragments, meaning every piece of plastic ever produced still exists in some form, and the total mass of microplastics in the ocean is growing exponentially
- Ocean gyres act as one-way traps: currents sweep plastic inward but the calm gyre center has no mechanism to flush it out, creating permanent and growing accumulation zones that function as toxic reservoirs
- Biomagnification through marine food webs means that a zooplankton consuming microplastics at parts-per-billion concentrations can produce toxin levels of parts-per-million in tuna — a million-fold concentration increase affecting human seafood consumers

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Plastic biodegrades in the ocean"	Plastic does not biodegrade in marine environments — it photodegrades. UV radiation breaks polymer chains, and mechanical forces fragment the weakened material into progressively smaller pieces. But the polymer molecules themselves persist. A plastic bottle does not become water and CO2 like a leaf does; it becomes billions of microplastic fragments that retain their chemical composition for centuries. This distinction between fragmentation and decomposition is central to understanding why the problem grows worse even as individual items seem to 'disappear.'
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Ocean cleanup technology can solve the problem"	Current cleanup technology is real and valuable for removing large debris from concentrated areas, but it faces a fundamental scale mismatch. The most ambitious project (The Ocean Cleanup) aims to remove 10,000 tons per year — but 11 million tons enter annually. That's removing 0.09% of annual input. Furthermore, as plastic fragments below 1mm, it becomes impossible to filter from water without also capturing plankton. The model shows that cleanup is necessary but can never be the primary solution — source reduction must do the heavy lifting.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Microplastics are only a problem for marine life, not humans"	Microplastics have been detected in human blood, lungs, placenta, and breast milk. The average person ingests approximately 5 grams of microplastic per week through food, water, and air. Laboratory studies link microplastic exposure to inflammation, endocrine disruption, and cellular damage. While long-term human health effects are still being studied, the precautionary principle applies — we are all part of the food web, and biomagnification means that marine contamination reaches our plates.